

CAPSTONE PROJECT SUBMISSION

EcoRoute: SDG 12 Agentic Logistics System

Project Concept Note

Innovator: **Kashmira Desai**
Institution: **Yashwantrao Bhonsale Institute of Technology, Sawantwadi**
Affiliation: **University of Mumbai**
Program: **Lenovo Leap Next Gen Scholar Program**
Focus Area: **UN SDG 12 (Responsible Consumption & Production)**

Project Concept Note

Introduction & The SDG 12 Challenge

Modern supply chains generate substantial environmental overhead, disproportionately affecting global sustainability indices. UN Sustainable Development Goal 12 (Responsible Consumption & Production) demands a systemic shift toward waste mitigation and resource optimization. While large multinationals deploy capital-intensive infrastructure to limit environmental harm, Small and Medium Enterprises (SMEs) face high economic and operational boundaries that exclude them from standard carbon and waste management platforms [cite: 33, 38].

The Last-Mile Sustainability Paradox: Last-mile delivery represents the most fragmented, carbon-intensive, and resource-inefficient segment of logistics [cite: 36]. Driven by partial vehicle loads, dynamic urban traffic constraints, and repeated delivery attempts, last-mile execution silently compounds fuel expenditures and per-parcel greenhouse gas (GHG) emissions [cite: 4, 36].

Core Architectural Philosophy: Agentic AI vs. Static Automation

Traditional logistics automation relies on rigid scripts characterized by fixed *if/then* pathways and hardcoded decision rules [cite: 41, 42]. These solutions yield predictable errors when exposed to real-world operational variations, requiring intensive manual troubleshooting and continuous engineering overhead [cite: 44, 46].

EcoRoute overcomes these restrictions by implementing an autonomous agentic framework in Python [cite: 7]. Unlike static scripts, an agentic configuration natively evaluates open-ended constraints, dynamically routes multi-stop configurations, manages real-time material inventory degradation vectors, and leverages external APIs to optimize performance on the fly [cite: 48, 50, 51].

Technical Workflow & Component Stack

The backend engine models logistics tasks across specialized AI personas that collaborate natively to balance efficiency and sustainability metrics [cite: 47, 48]:

- **Supply Chain Waste Analyzer:** Evaluates incoming cargo configuration matrix parameters (e.g., mass, transit duration, item shelf-life) to flag material waste threats and shelf-life decay factors under SDG 12 constraints [cite: 4, 32, 33].
- **Green Logistics Route Optimizer:** Processes the asset footprint to build an eco-friendly transport layout [cite: 7]. It computes emission reduction metrics and evaluates alternative fuel utilization targets to optimize deliveries [cite: 15].

The front-end user experience is driven by a lightweight Streamlit dashboard securely linked to production cloud instances via automated API credential orchestration, ensuring operational data privacy and seamless user interactions.